FEDERAL SPECIFICATION

LEATHER, CATTLEHIDE (INSOLE)

This amendment, which forms a part of Federal Specification KK-L-159, dated December 6, 1956, was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

Page 1, paragraph 1.2.1: Delete and substitute new paragraph 1.2.1, as follows:

1.2.1 *Types.*—Insole leather, in the form of cut incoles, covered by this specification, shall be furnished in the following types and tannages, as specified in the invitation for bids:

Type I.—Combination tanned.

Tannage a.—Chrome-retanned.

Tannage b.—Vegetable-retanned.

Type II.—Straight vegetable.

Page 2, paragraph 3.1.1: Delete and substitute new paragraph 3.1.1, as follows:

3.1.1 *Type I.*—The leather for type I shall be cattlehide tanned by one of the following methods:

Tannage a.—Tanned by a chrome tanning process followed by a vegetable retannage.

Tannage b.—Tanned by a vegetable tanning process followed by a chrome retannage.

Page 3, paragraph 3.7.1, table II: Delete and substitute the following table:

TABLE II.—*Chemical requirements, moisture-free basis*

Characteristics	Minimum	Maximum
Degree of tannage	35.0	---
Water solubles, percent	---	26.0
Grease (chloroform extract), percent	1.8	10.0
Acidity, pH	3.0	4.0
Paranitrophenol, percent	0.18	0.40
Chrome, Cr_2O_3, percent	2.0	---

Page 4, following paragraph 4.3.4 add new paragraph 4.3.4.1:

4.3.4.1 *Correction for nitrogenous additives.*—When a finishing treatment containing nitrogenous additives is used in the treatment of the vegetable-tanned type II leather, the supplier of insole leather shall submit for each lot a certification of the percent increase in nitrogen content (dry leather basis) resulting from the treatment. The certified percent increase in nitrogen shall be deducted from the total nitrogen found by analysis. The difference shall be used as the corrected nitrogen content for calculating hide substance and degree of tannage.

Page 5, paragraph 6.1: Delete and substitute new paragraph 6.1, as follows:

6.1 Ordering data. — Purchasers should specify the type, tannage, class, and thickness required and should exercise any desired options offered herein (see 1.2, 3.2, 5.1, 5.2.1.1, 5.2.2.1). Military agencies should also specify whether the insoles shall be packaged and packed for Level A—Military, Level B—Limited Military, or Level C—Minimum Military described as follows:

(1) *Level A—Military.*—Applies to that material to be transported by ocean-going or coastwise vessel and handled, issued and stored in the military supply system.

(2) *Level B—Limited Military.*—Applies to that material for storage and reshipment to a domestic receiving activity not involving sea transportation.

(3) *Level C—Minimum Military.* — Applies to that material required for immediate use without storage or reshipment by the initial domestic receiving activity.

MILITARY INTERESTS:
Army—M Q
Navy—S MC
Air Force

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